

Purification RFP in Cirac–Perez–Garcia–Schuch–Verstraete 2017

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1 Source statement

Cirac–Pérez–Garcia–Schuch–Verstraete define a purification renormalization fixed point for an MPDO tensor M through a purifying MPS tensor A :

$$\rho_p^{(N)}(M) = \text{tr}_a(|\Psi^{(N)}(A)\rangle\langle\Psi^{(N)}(A)|).$$

The source definition says that M is a PRFP when it admits such a representation with A an RFP.¹

The following theorem in the source states that, for tensors M satisfying the purification equation above, the three conditions

M is a PRFP, M has ZCL, $\rho^{(N)}(M)$ has the displayed PRFP decomposition

are equivalent.² The later warning that this definition is too weak concerns saturation of the area law, not the implication from PRFP to zero correlation length.

2 Deleted formalization

The deleted predicate `MPOTensor.IsPRFP` did not formalize the source definition. It did require a local LPDO witness and the condition

$$\text{MPSTensor.IsRFP}(\text{purifyingMPSTensor}(A)),$$

so the pure-state RFP requirement itself was present. The missing source data was the stronger purification-RFP structure around `eq:MPDO-Puri-1`: after tracing the ancillary index, the source obtains a trace-preserving completely positive map on the spin degrees of freedom and the decomposition displayed in `PuriRFP-PuriRFP-BNT`. The purifying tensor is also described by the canonical

¹Source location: `Papers/1606.00608/MPDO-22-12-17-2.tex:750–763`, Definition `def:Puri-RFP`.

²Source location: `Papers/1606.00608/MPDO-22-12-17-2.tex:775–786`.

RFP form `III_CFI_RFP`, but this is not the condition violated by the deleted scalar witness; in bond dimension 1 the witness already has that form. The deleted predicate kept only the local purifying family together with generic transfer-map idempotence of the doubled MPS tensor. It did not encode the global purification equation or the trace-preserving post-ancilla map.

With this weaker hypothesis the development proved a counterexample to PRFP implying ZCL. That conclusion contradicted the source theorem above, so it was a counterexample to the local Lean predicate, not to Definition `def:Puri-RFP`.

For this reason the file `TNLean/MPS/MPDO/PRFP.lean`, the corresponding blueprint section, and the counterexample registry entry were removed. The coverage audit now records the source PRFP definition as open, not as formalized.³

3 Faithful replacement

A faithful formalization should first define the mixed-state RFP condition in the source’s trace-preserving-map language. It should then state the purification RFP definition with the global purification equation `eq:MPDO-Puri-1` and the induced trace-preserving completely positive map appearing after the ancillary trace, recording the canonical RFP form `III_CFI_RFP` as part of the source structure. Only after that definition is in place should the theorem

$$\text{PRFP}(M) \iff \text{ZCL}(M)$$

be attached to the source theorem.

No theorem should assert that source PRFP fails to imply ZCL. Any future counterexample must instead be phrased for a strictly weaker local-purification predicate and must not be cited as Definition `def:Puri-RFP`.

References

³This note records the divergence removed in PR #2386. The faithful replacement is tracked by issue #2383 and the MPDO coverage tracker issue #2380.